

Module: 2

Algebraic Structures

Semigroups and Monoids

Introduction:

Let A be any set.

A mapping $f: A \times A \times A \dots \times A \rightarrow A$

(or) $f: A^n \rightarrow A$ is called n -ary operation

where ' n ' is called the order of the operation.

When $n=0$, 0-ary operation by means of particular distinguished element.

When $n=1$, unary operation.

When $n=2$, binary operation. {uses the following notation $*$, Δ , $+$, $\oplus$, etc.}

Algebraic system:

A set with one or more n -ary operation is called Algebraic system. (or) Simply Algebra.

denote $(A, f_1, f_2, \dots)$ where A - set, f_i are operations on A .

Example of Algebraic system:

- (1) Semigroups (2) Monoids (3) Groups. ($n=1$)
- (4) Rings, fields, Integral domain, ... etc, ($n=2$).

General Properties of binary operation

(i) Closure property:

Closure property:
A binary operation: $*$: $A \times A \rightarrow A$ is said to be closed if for all $a, b \in A$, the element $a * b \in A$.

Example: 1 Let $A = \mathbb{Z}$ { set of all integers }

* $I \times I \rightarrow I$ by $a * b = a + b$
 (or) $a * b = a \times b$
 (or) $a * b = a - b$
 (or) $a * b = a^b$ } one closed.

but $a * b = \frac{a}{b}$ is not closed $\because \frac{2}{3} \notin \mathbb{Z}$

(2) Associative property:

For all $a, b, c \in G$,

$$a * (b * c) = (a * b) * c$$

Ex.2: Let $G = \mathbb{Z}$, $*$: $G^2 \rightarrow G$ by $a * b = a + b$ is associative

but $a \times b = a - b$ is not associative.

$$\therefore (a * b) * c = (a - b) * c = a - b - c$$

$$\begin{aligned} \therefore (a * b) * c &= (a - b) * c = a - c \\ \text{bwr } a * (b * c) &= a * (b - c) = a - (b - c) = a - b + c \end{aligned}$$

(3) Let $*$ be a binary operation on a set G . If there exist an element $e_1 \in G$ s.t. $e_1 * a = a \quad \forall a \in G$, then e_1 - left identity w.r. to $*$. Similarly if there exist an element $e_2 \in G$ s.t. $a * e_2 = a \quad \forall a \in G$ then e_2 - right identity of G w.r. to $*$.

If there exist an element $e \in G$ s.t.
 $a * e = e * a = a$ then we call $e \in G$ is an
identity element of G .

It's also called an distinguished element.

Example:-

Ex 3 Let $G = \mathbb{Z}$, $* : G^2 \rightarrow G$ by

(i) $a * b = a + b$ here $e = 0$ $\therefore 0 + a = a + 0 = a$

(ii) $a * b = a \times b$ here $e = 1$ $\therefore a \times 1 = 1 \times a = a$

(iii) $a * b = a + b - 2$ here $e = 2$ $\therefore \begin{cases} a + e = a + 2 - 2 = a \\ e + a = 2 + a - 2 = a \end{cases}$

Ex 4: (iv) $a * b = a + b + 2ab$ here $e = 0$

$\therefore a * e = a + 0 + 0 = a$
 $e * a = 0 + a + 0 = a$

Ex 4: Let $G = \mathbb{Q}^+$, Let $*$ defined by $a * b = \frac{ab}{2}$

$\therefore a * e = \frac{ae}{2} = a \Rightarrow \boxed{e = 2}$

$e * a = \frac{ea}{2} = a \Rightarrow \boxed{e = 2}$

Then

Ex 5: $G = \{1, -1, i, -i\}$ where $*$ defined by

$a * b = ab$

Here $e = 1$ $\therefore a * e = a = e * a$

(4) Commutative property:

For all $a, b \in G$, if $a * b = b * a$ then $*$ is commutative.

Example:

Ex 1: Let $G = \mathbb{R}$ where $*$ defined by

(i) $a * b = a + b$

is commutative $[\because a + b = b + a]$

(ii) $a * b = ab$

is commutative $[\because ab = ba]$

(iii) $a * b = a - b$

is not commutative
 $[\because a - b \neq b - a]$

(iv) $a * b = \frac{a}{b}$

is not commutative
 $\therefore [\frac{a}{b} \neq \frac{b}{a}]$

(5) Distributive property

Let $*$ and Δ be two binary operation on a set A
if for all $a, b, c \in A$

- (i) $a * (b \Delta c) = (a * b) \Delta (a * c)$ { left distributive }
& (ii) $(b \Delta c) * a = (b * a) \Delta (c * a)$ { Right distributive }

Ex: 1 $(\mathbb{Z}, +, \times)$ is distributive

$$\therefore (i) a + (b \times c) = (a + b) \times (a + c)$$

$$& (ii) (b \times c) + a = (b + a) \times (c + a)$$

(or)

$$(i) a \times (b + c) = (a \times b) + (a \times c)$$

$$(ii) (b + c) \times a = (b \times a) + (c \times a)$$

(6) Cancellation law:

A binary operation denoted by $*$ in a set A
is said to satisfy,

(i) Left cancellation law

if for all $a, b, c \in A$, $a * b = a * c \Rightarrow b = c$

(ii) Right cancellation law.

if for all $a, b, c \in A$, $b * a = c * a \Rightarrow b = c$

(1) Inverse property:

Let $*$ be binary operation on a non-empty set A and e be the identity element of A w.r to $*$. An element $a \in A$ is said to be invertible if there exist an element $b = a^{-1} \in A$

$$\text{s.t. } a * b = b * a = e$$
$$\text{[i.e. } a * a^{-1} = a^{-1} * a = e]$$

Here we say a and b are inverse to each other.

Ex: 1: Let $Z = \mathbb{Z}$, $*$ defined by

$$(i) \quad a * b = a + b$$

Here inverse is exist for all element in Z i.e for an element $z \in Z$, the inverse of z is $-z \in Z$.

$$\therefore z + (-z) = 0 = (-z) + z \quad \begin{matrix} \text{[0 - identity]} \\ \text{w.r to +} \end{matrix}$$

$$(ii) \quad a * b = a \cdot b \quad [1 \text{ is identity w.r to } \cdot]$$

$\therefore Z$ has inverse for 1 and -1

$$\text{where inverse of } 1 \text{ is } 1 \quad \because 1 * 1 = 1 \cdot 1 = 1$$
$$-1 \text{ is } -1 \quad \because -1 * -1 = 1$$

For other element inverse not exist.

Theorem: 1 The identity element of a binary operation $*$ on a set A , if it exists and is unique.

Proof: If possible, let e_1 and e_2 be two identity elements on A w.r.t to $*$

$$\text{Then } \forall a \in A, a * e_1 = e_1 * a = a \quad \text{--- (1)}$$

$$\& a * e_2 = e_2 * a = a \quad \text{--- (2)}$$

$$\text{As } e_1 \in A, (2) \Rightarrow e_1 * e_2 = e_2 * e_1 = e_1$$

$$e_2 \in A, (1) \Rightarrow e_2 * e_1 = e_1 * e_2 = e_2$$

$$\Rightarrow \boxed{e_1 = e_2}$$

Some Examples:

(i) Consider a finite set S and its power set $P(S)$
Then the Algebraic system $[P(S), \cup, \cap]$ satisfies the following property.

(i) For any $A, B \in P(S)$, $A \cup B \in P(S)$ - Closure w.r.t $\cup$

(ii) For any $A, B \in P(S)$, $A \cap B \in P(S)$ - Closure w.r.t $\cap$

(iii) For any $A, B, C \in P(S)$

$$A \cup (B \cap C) = (A \cup B) \cap C \quad \& \quad A \cap (B \cup C) = (A \cap B) \cup C$$

$\therefore$ associative under $\cup$ and $\cap$

(iv) For any $A, B \in P(S)$

$$A \cup B = B \cup A \quad \& \quad A \cap B = B \cap A$$

Commutative under $\cup$ & $\cap$

(v) For $A, B, C \in P(S)$

$$A \cup (B \cap C) = (A \cup B) \cap C \quad \& \quad A \cap (B \cup C) = (A \cap B) \cup C$$

$\therefore$ distributive law is true under $\cup$ & $\cap$.

(vi) The element $\phi \in P(S)$ is an identity element with respect to $\cup$. $\therefore A \cup \phi = \phi \cup A = A \quad \forall A \in P(S)$

& the element $S \in P(S)$ is an identity element w.r.to $\cap$. $\therefore A \cap S = S \cap A = A \quad \forall A \in P(S)$

$\therefore$ Identity element exist w.r.to $\cup$ & $\cap$.

(vii) For Inverse
under $\cup$. $\therefore e = \phi$ is identity w.r.to $\cup$

$\therefore$ For any $A \in P(S)$, find $B \in P(S)$ s.t $A \cup B = \phi$
Such B cannot exist always

except when $A = \phi, B = \phi$.

$\therefore$ Inverse not exist for all $A \in P(S)$ except ϕ .

under $\cap$: $\therefore e = S$ is identity elt w.r.to $\cap$

$\therefore$ For any $A \in P(S)$, find $B \in P(S)$ s.t $A \cap B = S$

$\therefore$ Such B cannot exist always except $A = S, B = S$.

$\therefore$ Inverse not exist always except $A = S$.

(viii) Cancellation law is also not true always

$$\begin{aligned} \therefore A \cup B = A \cup C &\not\Rightarrow B = C \\ \& A \cap B = A \cap C &\not\Rightarrow B = C \end{aligned}$$

(2) Consider the set R : Real numbers together with operation $+$ and $\cdot$ (usual addition and multiplication)

$(R, +, \cdot)$ - Satisfies all the property (i) $\rightarrow$ (iv)

where $+$ defined as $a+b$ for all $a, b \in R$
 $\cdot$ defined as $a \cdot b$ for all $a, b \in R$.

(3) Consider $B = \{1, -1, i, -i\}$ and an operation $*$ on B with the following table.

x	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	1	-1
-i	-i	i	-1	1

TABLE 1

The Algebraic System

$(B, *)$ satisfies

all the properties (i) $\rightarrow$ (vii)

The identity element w.r to $*$ is 1

Inverses of 1 is 1, -1 is -1, i is -i and -i is i.

Note: The above table is called the Cayley representation table.

[It will be defined if the number of elements in a set is finite].

(4) Consider the set $B = \{0, 1\}$ and the operation $+$ on B is given by the following table.

x	0	1
0	0	1
1	1	0

Table 2

$+$	0	1
0	0	0
1	0	1

Table 3

The System $(B, +)$ satisfies the property.

(i) to (vii)

(5) Let $X = \{a, b\}$ and S denote the set of all mappings from X to X . Let $S = \{f_1, f_2, f_3, f_4\}$

where $f_1(a) = a$ | $f_1(b) = b$ | $f_2(a) = a$ | $f_2(b) = a$ | $f_3(a) = b$ | $f_3(b) = b$ | $f_4(a) = b$ | $f_4(b) = a$

Then $\langle S, \circ \rangle$, where $\circ$ denote the composition of functions is an algebraic system.

f_1 - identity element w.r to $\circ$.

The Cayley representation table of $(S, \circ)$

$\circ$	f_1	f_2	f_3	f_4
f_1	f_1	f_2	f_3	f_4
f_2	f_2	f_2	f_2	f_2
f_3	f_3	f_3	f_3	f_3
f_4	f_4	f_3	f_2	f_1

Table 4

Ex

$$f_2 \circ f_4 =$$

$$f_2 \circ f_4(a) = f_2(f_4(a)) = f_2(b) = a = f_2(a)$$

$$f_2 \circ f_4(b) = f_2(f_4(b)) = f_2(a) = a = f_2(b)$$

$$\therefore f_2 \circ f_4 = f_2$$

Example.

$$f_4 \circ f_2 =$$

$$f_4 \circ f_2(a) = f_4(f_2(a)) = f_4(a) = b \rightarrow f_3$$

$$f_4 \circ f_2(b) = f_4(f_2(b)) = f_4(a) = b$$

(6) Let $X = \{1, 2, 3, 4\}$ and $f: X \rightarrow X$ by
 $f = \{ \langle 1, 2 \rangle, \langle 2, 3 \rangle, \langle 3, 4 \rangle, \langle 4, 1 \rangle \}$
 i.e., $f(1) = 2, f(2) = 3, f(3) = 4, f(4) = 1$.

Let f^0 be an identity mapping from X to X
 i.e., $f^0(1) = 1, f^0(2) = 2, f^0(3) = 3, f^0(4) = 4$.

and let $f^2 = f \circ f, f^3 = f^2 \circ f, f^4 = f^3 \circ f$ and so on.

Consider the set $F = \{f^0, f^1, f^2, f^3, f^4\}$.

$\therefore \langle F, \circ \rangle$ is an Algebraic System.

$\circ$	f^0	f^1	f^2	f^3
f^0	f^0	f^1	f^2	f^3
f^1	f^1	f^2	f^3	f^0
f^2	f^2	f^3	f^0	f^1
f^3	f^3	f^0	f^1	f^2

Table 5

⑦ Let $Z_n = \{[0], [1], \dots, [n-1]\}$ set of residue classes of 'n'

Here $[j]$ denote set of all those integers which leaves a remainder j when it is divided by n .

The operation $+_n$ on Z_n given by

$$[i] +_n [j] = [(i+j) \pmod n]$$

for all $i, j = 0, 1, 2, \dots, n-1$.

The system $\langle Z_4, +_4 \rangle$ is algebraic system. which satisfies closure, Associative, Commutative distributive and so on.

The Cayley Representation table.

$+_4$	$[0]$	$[1]$	$[2]$	$[3]$
$[0]$	0	1	2	3
$[1]$	1	2	3	0
$[2]$	2	3	0	1
$[3]$	3	0	1	2

Table 5

Table 6

Observation: From Table 5 and Table 6, we see there is a similarity between.

Its obtained one from other by simply relabeling f^0, f^1, f^2, f^3 by $[0], [1], [2], [3]$.

and the operation $\circ$ to $+_4$

Let us define

$$\psi: F \rightarrow Z_4 \text{ by } \psi(f^i) = [i]$$

for $i = 0, 1, 2, 3$.

From table it is clear that

$$\psi(f^i \circ f^j) = \psi(f^i) +_4 \psi(f^j) \quad \text{--- (2)}$$

for all $i, j = 0, 1, 2, 3$.

(2) shows that the image of ψ for the argument $f^i \circ f^j$ is same as the result of the operation $+_4$ applied to the image. $\psi(f^i) = [i]$ and $\psi(f^j) = [j]$ of the elements f^i and f^j .

$\Rightarrow$ The mapping preserves the operation $\circ$ and $+_4$.

$$\text{ie, } \begin{array}{ccc} F \times F & \xrightarrow{\circ} & F \\ \downarrow \psi \times \psi & & \downarrow \psi \\ \mathbb{Z}_4 \times \mathbb{Z}_4 & \xrightarrow{+_4} & \mathbb{Z}_4 \end{array}$$

This shows that the Algebraic System $\langle F, \circ \rangle$ and $\langle \mathbb{Z}_4, +_4 \rangle$ are structurally same. We say $\langle F, \circ \rangle$ is homomorphic to $\langle \mathbb{Z}_4, +_4 \rangle$.

Defn: Homomorphism

Let $\langle X, \circ \rangle$ and $\langle Y, * \rangle$ be two AS (Algebraic System) of the same type ($\circ$ and $*$ are binary operation).

A mapping $g: X \rightarrow Y$ is called homomorphism (or) morphism from $\langle X, \circ \rangle$ to $\langle Y, * \rangle$ if for

any $x_1, x_2 \in X$

$$g(x_1 \circ x_2) = g(x_1) * g(x_2)$$

If such g exist between X and Y then we say $\langle Y, * \rangle$ is an homomorphic image of $\langle X, \circ \rangle$.

Defn: Isomorphism

Let g be a homomorphism from $\langle X, \circ \rangle$ and $\langle Y, * \rangle$. If $g: X \rightarrow Y$ is onto then g is called epimorphism. If $g: X \rightarrow Y$ is one-one then g is called monomorphism.

If $g: X \rightarrow Y$ is both one-one and onto (bijective) then we say g is isomorphism.

If such g exist between any two Algebraic system say X and Y then we say X isomorphic to Y (or) $X \cong Y$.

Example: 1 $\psi: F \rightarrow Z_4$

$$\text{by } \psi(f^i) = [i], i = 0, 1, 2, 3.$$

ψ is homomorphism, one-one & onto

$\Rightarrow \psi$ is isomorphism between F & Z_4 .

note: If g is an isomorphism between X to X then we say g is automorphism.

Example: 1: $f: (\mathbb{C}, +) \rightarrow (\mathbb{C}, +)$

$$f(a+ib) = a-ib$$

check f is homo: $f(z_1 + z_2) = f(z_1) + f(z_2)$

$$\text{Let } z_1 = a_1 + ib_1 \text{ \& } z_2 = a_2 + ib_2$$

$$\begin{aligned} \Rightarrow f(z_1 + z_2) &= f(a_1 + ib_1 + a_2 + ib_2) \\ &= f(a_1 + a_2 + i(b_1 + b_2)) \\ &= (a_1 + a_2) - i(b_1 + b_2) \end{aligned}$$

$$f(z_1) + f(z_2) = a_1 - ib_1 + a_2 - ib_2 = (a_1 + a_2) - i(b_1 + b_2)$$

$$\Rightarrow f \text{ is homo.}$$

(ii) f is one-one
ie, to check if $f(z_1) = f(z_2) \Rightarrow z_1 = z_2$

$$\text{consider } f(z_1) = f(z_2)$$

$$\Rightarrow a_1 - ib_1 = a_2 - ib_2$$

$$\Rightarrow a_1 = a_2 \text{ \& } b_1 = b_2$$

$$\therefore z_1 = z_2$$

(iii) f is onto

$\forall z \in \mathbb{C} \exists z' \in \mathbb{C} \text{ s.t. } f(z') = z$

$$\text{Let } z = a + ib$$

$$\text{Then consider } z' = a - ib$$

$$\Rightarrow f(z') = f(a - ib) = a + ib$$

$\therefore f$ is onto

$\Rightarrow f$ is automorphism.

Some other example

(1) Let $S = \{a, b, c\}$ w.r to $*$
& $P = \{1, 2, 3\}$ w.r to $\oplus$

Their Cayley tables

$*$	a	b	c
a	a	b	c
b	b	b	c
c	c	b	c

$\oplus$	1	2	3
1	1	2	1
2	1	2	2
3	1	2	3

Consider the map: $g: S \rightarrow P$ by

$$g(a) = 3, g(b) = 1, g(c) = 2$$

using Cayley table, g is one-one, onto
and homomorphism

$\Rightarrow g$ is isomorphism.

$$\left[\begin{array}{l} \therefore g(a * c) = g(a) \oplus g(c) \\ \text{d with} \\ \text{anner} \end{array} \right. \left. \begin{array}{l} g(c) = 3 \oplus 2 \\ 2 = 2 \end{array} \right| \begin{array}{l} g(b * c) = g(b) \oplus g(c) \\ g(c) = 1 \oplus 2 \\ 2 = 2 \end{array}$$

(2) Let $\phi: (\mathbb{Z}, +) \rightarrow (\mathbb{Z}\mathbb{Z}, +)$

by $\phi(x) = 2x$ is homomorphism.

$$\phi(x+y) = \phi(x) + \phi(y)$$

$$[\because \phi(x+y) = 2(x+y) = 2x + 2y = \phi(x) + \phi(y)]$$

ϕ is one-one and onto

$\therefore \phi$ is isomorphism

$$\mathbb{Z} \cong 2\mathbb{Z}$$

(3) Let $g: \langle \mathbb{N}, + \rangle \rightarrow \langle \mathbb{Z}_4, +_4 \rangle$

by $g(x) = [x \pmod{4}]$ for any $x \in \mathbb{N}$

Note $g(x+y) = [(x+y) \pmod{4}]$

$$= [x] +_4 [y] = g(x) +_4 g(y)$$

$\Rightarrow g$ is homomorphism

Note g is onto

$\therefore$ for every element $a \in \mathbb{Z}_4$

there exist an element $x \in \mathbb{N}$

$$\text{s.t. } g(x) = [x] = a$$

but g is not one-one.

if

$$\therefore g(x) = g(y)$$

$$[x] = [y]$$

$$\Rightarrow x \neq y$$

$\therefore [x]$ set of all elt
of $\mathbb{N}$ which leave
the remainder x .

(4) prove that $f: S \rightarrow T$ is a homomorphism from $(S, *)$ to (T, Δ) and $g: T \rightarrow P$ is a homomorphism from (T, Δ) to (P, ∇) . Then $g \circ f: S \rightarrow P$ is a homomorphism from $(S, *)$ to (P, ∇) .

Ans:

Given: $f: S \rightarrow T$ is homo

$$\Rightarrow f(s, * s_2) = f(s_1) \Delta f(s_2) \quad \text{--- ①}$$

& $g: T \rightarrow P$ is homo

$$g(t, \Delta t_2) = g(t_1) \nabla g(t_2) \quad \text{--- ②}$$

$\Rightarrow$ T.P.T $g \circ f: S \rightarrow P$ is homo.

enough to prove: $g \circ f(s, * s_2) = g \circ f(s_1) \nabla g \circ f(s_2)$

$$\begin{aligned} \text{L.H.S.} \quad g \circ f(s, * s_2) &= g(f(s, * s_2)) && \text{use (1)} \\ &= g(f(s_1) \Delta f(s_2)) && \text{use (1)} \\ &= g(f(s_1)) \nabla g(f(s_2)) && \because f(s_1), f(s_2) \in T \\ &= g \circ f(s_1) \nabla g \circ f(s_2) \\ &= \text{R.H.S.} \end{aligned}$$

Semigroup:

Let G be a non-empty set and $*$ be a binary operation defined on G .

The AS $(G, *)$ is called semigroup if the operation $*$ is Associative.

$$\text{ie, } \forall a, b, c \in G, a * (b * c) = (a * b) * c$$

[Observation: $*$ also posses the closure property].

Monoid: A semigroup $(G, *)$ with an identity element with respect to $*$ is called Monoid.

ie, if $(G, *)$ is monoid then

- (i) $*$ is associative.
- (ii) $\exists$ an elt $e \in G$ such that
 $\forall a \in G, a * e = e * a = a$

Group: A monoid $(G, *)$ with an inverse element for every element of G is called group.

(or)

$(G, *)$ is group. then $*$ satisfies.

(i) Associative

(ii) identity

(iii) Inverse

ie, $\forall a \in G, \exists b \in G, (b = a^{-1})$

$$\text{s.t. } a * a^{-1} = a^{-1} * a = e.$$

Abelian group: A group $(G, *)$ is said to be abelian if $*$ also satisfies commutative property.

Idempotent element: An element $a \in G$, where $(G, +)$ is a S is said to be idempotent then $a * a = a$.

Cyclic group: A monoid $(G, *, e)$ is said to be cyclic if there exist an element $a \in G$ such that for every element of G can be written as some powers of a .

ie, $\forall b \in G$,

$$b = a^i \text{ for some } i \in \mathbb{N}$$

$$\text{ie, } b = a * a * \dots * a \text{ (i times)}$$

Example:- Let $F = \{1, -1, i, -i\}$ with usual multiplication is cyclic.

Here generator is i .

$$1 = i^4 = i \cdot i \cdot i \cdot i$$

$$-1 = i^2 = i \cdot i$$

$$i = i^1 = i$$

$$-i = i^3 = i \cdot i \cdot i$$

(or) $-i$ is also a generator.

Note: If $(G, *)$ is a cyclic group with generator say a then a^{-1} is also a generator for $(G, *)$.

Order of an element:

Let $\langle G, * \rangle$ be a group. Let $a \in G$,
with $a^n = e$, where n is the least positive
int satisfies above. Then we say n is the
order of a . (or) $o(a) = n$.

Idempotent element

Let $\langle G, * \rangle$ be an AS. An element $a \in G$
is said to be idempotent then $a * a = a$
i.e., $a^2 = a$.

Examples of semigroup, monoid, group.

- ① Let Z be the set of all integers and
 $*$ is a binary operation defined on Z
by $a * b = a + b - ab$. Determine whether
 $(Z, *)$ is monoid. Is $(Z, *)$ is group?
Also check $*$ satisfies commutative.

Ans:

Given: Z : Set of all integers.
 $*$: $Z^2 \rightarrow Z$ by $a * b = a + b - ab$ — ①

(i) Closure:
 $\forall a, b \in Z$, $a * b = a + b - ab \in Z$
 $\therefore *$ is closed.

(ii) Associative:
 $\forall a, b, c \in Z$, TPT $(a * b) * c = (a * (b * c))$
LHS, $(a * b) * c = (a + b - ab) * c$
 $= a + b - ab + c - (a + b - ab)c$
 $= a + b + c - ab - ac - bc + abc$

$$\begin{aligned}
 \text{RHS: } a * (b + c) &= a + (b + c - bc) \quad \text{use (r)} \\
 &= a + b + c - bc - a(b + c - bc) \\
 &= a + b + c - bc - ab - ac + abc
 \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore *$ is Associative

(iii) identity element:

If $e \in \mathbb{Z}$ is an identity then it satisfies
 $a * e = e * a = a \quad \forall a \in \mathbb{Z}$

$$\text{Consider } a * e = a \quad \text{use (i)}$$

$$a + e - ae = a$$

$$e(1 - a) = 0$$

$$e = \frac{0}{1-a} = 0 \quad \text{provided } a \neq 1$$

$\therefore e = 0$ is an identity element provided $a \neq 1$

$\therefore (\mathbb{Z}, *)$ is a monoid.

Now check $(\mathbb{Z}, *)$ is a group.

(iv) Inverse:

$\forall a \in \mathbb{Z}$, we need to find $a^{-1} \in \mathbb{Z}$ s.t.

$$a * a^{-1} = e = 0, \quad a \neq 1$$

$$\Rightarrow a + a^{-1} - aa^{-1} = 0$$

$$a^{-1}(1 - a) = -a$$

$$a^{-1} = \frac{-a}{1-a}, \quad a \neq 1$$

$$\text{Note } a^{-1} = \frac{-a}{1-a} \notin \mathbb{Z} \text{ for all } a$$

$\therefore$ Inverse is not exist in $\mathbb{Z}$ w.r.t $*$
 $(\mathbb{Z}, *)$ is not group under $*$.

now to check commutative w.r.to $*$

ie; to prove for all $a, b \in \mathbb{Z}$

$$\Rightarrow a * b = b * a$$

$$a * b = a + b - ab = b + a - ba = b * a$$

$\therefore *$ is commutative.

② Let $\mathbb{Q}^+$ be a set of all positive rational numbers: $*$ is a binary operation defined on $\mathbb{Q}^+$ as $a * b = \frac{ab}{2}, \forall a, b \in \mathbb{Q}^+$.

(i) check $(\mathbb{Q}^+, *)$ is group.

(ii) check $(\mathbb{Q}^+, *)$ is Abelian.

Ans:

Given: $\mathbb{Q}^+ =$ set of all +ve rationals
 $*$ is $\mathbb{Q}^+ \times \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ by $a * b = \frac{ab}{2}, \forall a, b \in \mathbb{Q}^+$

(i) Closure:
 $\forall a, b \in \mathbb{Q}^+, a * b = \frac{ab}{2} \in \mathbb{Q}^+$

$\therefore *$ is closed.

(ii) Associative:

$$\forall a, b, c \in \mathbb{Q}^+ \quad \text{TPJ} \quad a * (b * c) = (a * b) * c$$

$$\text{LHS: } a * (b * c) = a * \left(\frac{bc}{2}\right) = \frac{a\left(\frac{bc}{2}\right)}{2} = \frac{abc}{4}$$

$$\text{RHS: } (a * b) * c = \left(\frac{ab}{2}\right) * c = \frac{\left(\frac{ab}{2}\right)c}{2} = \frac{abc}{4}$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore *$ is associative.

(iii) Identity element.

We need to find an elt $e \in \mathbb{Q}^+$
s.t $a * e = e * a = a \quad \forall a \in \mathbb{Q}^+$

Consider $a * e = a$

$$\Rightarrow \frac{a \cdot e}{2} = a \Rightarrow \boxed{e = 2}, a \neq 0$$

$\therefore e = 2$ is an identity elt of $\mathbb{Q}^+$
w.r to $*$

(iv) Inverse:

$\forall a \in \mathbb{Q}^+$ need to find $a^{-1} \in \mathbb{Q}^+$ s.t

$$a * a^{-1} = e = a^{-1} * a$$

Consider $a * a^{-1} = e = 2$

$$\Rightarrow \frac{a \cdot a^{-1}}{2} = 2$$

$$\boxed{a^{-1} = \frac{4}{a}}, a \neq 0$$

$$\therefore a^{-1} \in \mathbb{Q}^+$$

$\therefore \mathbb{Q}^+$ have inverse for all
element w.r to $*$

$\therefore (\mathbb{Q}^+, *)$ is a group.

Now to prove abelian:

it is enough to prove commutative
property

that is $\forall a, b \in \mathbb{Q}^+, a * b = b * a$

$$a * b = \frac{ab}{2} = \frac{ba}{2} = b * a$$

$\therefore *$ is commutative

$\therefore (\mathbb{Q}^+, *)$ is abelian.

- (3) prove that the set of all +ve integers under usual multiplication forms a monoid but not a group.

[$\therefore$ it has identity elt $\boxed{e=1}$ but $\forall a \in G, a^{-1} \notin G$]

- (4) Define a binary operation $*$ on the set R of all real numbers by $x*y = xy + x + y$. Determine $(R, *)$ is group or not. Also check $*$ is commutative or not. Find the idempotent element in $(R, *)$.

- (5) Let S be non-empty set. And let $P(S)$ be power set of S . Then prove that
- $(P(S), \cup)$ is monoid.
 - $(P(S), \cap)$ is monoid.
 - prove that $\cup, \cap$ are commutative.

- (6) Let $G = \{1, -1, i, -i\}$.
- Is G is a group under usual multiplication.
 - Is G is abelian under $*$.
 - Is G is cyclic under $*$.
 - find the order of every element in G under $*$.

Ans: The Cayley representation table

for $(G, *)$ is

$a \backslash b$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	1	-1
-i	-i	i	-1	1

From the table,

We can see $(\cdot)$ is closure, Associative.

The identity element of G under $\cdot$ is 1

$$\text{The inverse of } \begin{array}{l} (1) \text{ is } 1 \\ (-1) \text{ is } -1 \end{array} \quad \left| \begin{array}{l} \text{Inv}(i) = -i \\ \text{Inv}(-i) = i \end{array} \right.$$

$\therefore (G, \cdot)$ is group.

We can easily check that $\cdot$ is commutative.

$$[\because a \cdot b = b \cdot a]$$

$\therefore (G, \cdot)$ is abelian.

(iii) To prove $(G, \cdot)$ is cyclic.

We need to find an element $a \in G$ [generator] s.t. it will generate all element of G .

$$\text{Let } a = i. \text{ Note } \begin{array}{l} i^1 = i \\ i^2 = -1 \\ i^3 = -i \\ i^4 = 1 \end{array}$$

$\therefore (G, \cdot)$ is cyclic group with generator i [~~and~~ $-i$].

(iv) The order of an element: $a^n = e = 1$

$$\text{Note } \begin{array}{ll} (1)^1 = 1 & o(1) = 1 \\ (-1)^2 = 1 & o(-1) = 2 \\ (i)^4 = 1 & o(i) = 4 \\ (-i)^4 = 1 & o(-i) = 4 \end{array}$$

(1) Let $GL_n(\mathbb{R}) = \{ A : A \in M_{n \times n}(\mathbb{R}) \text{ and } \det(A) \neq 0 \}$

$GL_n(\mathbb{R}) =$ set of all $n \times n$ - invertible matrices.

check $(GL_n(\mathbb{R}), \text{matrix multiplication})$

form a group:

Is $(GL_n(\mathbb{R}) \text{ w.r.t. mat. mul.})$ is commutative.

Ans:

Given: $GL_n(\mathbb{R})$: set of all non-singular matrices.

binary operation is mat. mul. $\therefore A * B = AB$.

(i) Closure:

Note: $\forall A, B \in GL_n(\mathbb{R}), \det(A) \neq 0, \det(B) \neq 0$

$$AB = C_{n \times n} \text{ and } \det(C) = \det(AB) = \det(A)\det(B) \Rightarrow \det(C) \neq 0$$

$\therefore C \in GL_n(\mathbb{R})$.

$\therefore GL_n(\mathbb{R})$ is closed w.r.t. mat. mul.

(ii) Associative:

$\forall A, B, C \in GL_n(\mathbb{R})$

$$A(BC) = (AB)C$$

$\therefore$ mat. mul. is Associative.

(iii) Identity.

we need to find an elt E in $GL_n(\mathbb{R})$ s.t. $\forall A \in GL_n(\mathbb{R}), AE = EA = A$

it is true for $E = I_n$

$$\therefore \det(I_n) = 1 \neq 0$$

(iv) Inverse:

$\forall A \in GL_n(R)$, we need to find $A^{-1} \in GL_n$

$$\text{s.t. } AA^{-1} = I_n = A^{-1}A$$

Note for every $A \in GL_n(R)$,

$\therefore A$ is invertible, $\exists A^{-1} \in GL_n(R)$

$$\text{s.t. } AA^{-1} = A^{-1}A = I_n$$

$\therefore GL_n(R)$ is a group under mat. mul.

[Note: $GL_n(R)$ also called as general linear group].

Note that $AB \neq BA$

$\therefore$ mat. mul. is not commutative.

(8) Let G denote the set all matrices of the form $\begin{bmatrix} x & x \\ x & x \end{bmatrix}$, $x \in \mathbb{R}^+$. prove that G is group under mat. mul.

Ans:-

$$G = \left\{ \begin{bmatrix} x & x \\ x & x \end{bmatrix}, x \in \mathbb{R}^+ \right\} \quad * : \text{mat. mul.}$$

(i) Closure:

$$\text{Let } A = \begin{bmatrix} x & x \\ x & x \end{bmatrix}, B = \begin{bmatrix} y & y \\ y & y \end{bmatrix} \in G$$

$$AB = \begin{bmatrix} 2xy & 2xy \\ 2xy & 2xy \end{bmatrix} \in G \quad \because 2xy \in \mathbb{R}^+$$

$\therefore$ Closure is true.

Example of group:

Symmetric or permutation group:

Let S denote a set of n elements. Then
The set $S_n = \{ \text{set of all permutation on the set } S \}$

[note: a permutation p is a bijection between S to S]

Example: when $n=3$.

Let $S = \{a, b, c\}$

Then $S_n = \{ P_1 = \begin{pmatrix} a & b & c \\ a & b & c \end{pmatrix}, P_2 = \begin{pmatrix} a & b & c \\ b & a & c \end{pmatrix},$

$P_3 = \begin{pmatrix} a & b & c \\ c & b & a \end{pmatrix}, P_4 = \begin{pmatrix} a & b & c \\ a & c & b \end{pmatrix}$

$P_5 = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix}, P_6 = \begin{pmatrix} a & b & c \\ c & a & b \end{pmatrix} \}$

Let the binary operation $\diamond$ denote the composition of two permutations,

$$\diamond : S_n \times S_n \rightarrow S_n$$

by $P_i \diamond P_j = P_k$ for some i, j, k

Example: when $n=3$.

$$P_3 \diamond P_5 = \begin{pmatrix} a & b & c \\ c & b & a \end{pmatrix} \diamond \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix} = \begin{pmatrix} a & b & c \\ a & c & b \end{pmatrix} = P_4$$

The Cayley table for (S_3, Δ)

Δ	P_1	P_2	P_3	P_4	P_5	P_6
P_1	P_1	P_2	P_3	P_4	P_5	P_6
P_2	P_2	P_1	P_5	P_6	P_3	P_4
P_3	P_3	P_6	P_1	P_5	P_4	P_2
P_4	P_4	P_5	P_6	P_1	P_2	P_3
P_5	P_5	P_4	P_2	P_3	P_6	P_1
P_6	P_6	P_3	P_4	P_2	P_1	P_5

It is clear that (S_3, Δ) is a group.

The identity element is P_1 . Inverse exist for all element.

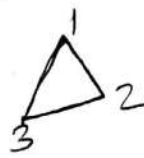
Note that (S_3, Δ) is not commutative

In general (S_n, Δ) is a group: (Symmetric group).
note $|S_n| = n!$

Dihedral group: Let $S = \{ \text{Set of all } n \text{ elements} \}$
Let $D_n = \{ \text{Set of all symmetries of the regular polygon on } n\text{-sides} \}$

Note when $n=3$: $D_3 = S_3$

$D_3 = \{ \text{rotation: } P_1, P_5, P_6 \}$
 $\{ \text{reflection: } P_2, P_3, P_4 \}$



rotation
 $P_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$, $P_6 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$
 $P_5 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$

about 1
 $P_4 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$

about 2
 $P_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$

reflection
 $P_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$



observe that (D_3, Δ) also form a group.

In general (D_n, Δ) is a group.

note order of $D_n = 2n$

[$\because$ There are n -rotation about $\frac{360}{n}$
& n -reflections]

Its clear that D_n forms a subset of S_n .

Subgroup: Let $(G, *)$ be a group
and let $S \subseteq G$ be such that it satisfies
the following.

(i) $e \in S$ where e is the identity of $(G, *)$

(ii) For any $a \in S$, $a^{-1} \in S$

(iii) For $a, b \in S$, $a * b \in S$

Then $(S, *)$ is called a subgroup of $(G, *)$.

ob: $(S, *)$ itself is a group.

Note that:

$\langle \{e\}, + \rangle$ and $\langle G, + \rangle$ are trivial subgroups of $\langle G, + \rangle$. All the other subgroups of $\langle G, + \rangle$ are proper subgroups.

Examples:

(1) $\langle D_n, \Delta \rangle$ is a subgroup of $\langle S_n, \Delta \rangle$.

(2) Find all the subgroups of $\langle \mathbb{Z}_5, +_5 \rangle$

Ans:

Given $\langle G, + \rangle = \langle \mathbb{Z}_5, +_5 \rangle$

Note $\mathbb{Z}_5 = \{[0], [1], [2], [3], [4]\}$

The Cayley table:

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Note

(i) $\langle \{[0]\}, +_5 \rangle$

and $\langle \mathbb{Z}_5, +_5 \rangle$

are trivial subgroups of $\langle \mathbb{Z}_5, +_5 \rangle$.

(ii) The proper subgroups:

There is no proper subgroup.

(3) Find all the subgroups of $\langle \mathbb{Z}_6, +_6 \rangle$

Note: $\langle \{[0]\}, +_6 \rangle$, $\langle \mathbb{Z}_6, +_6 \rangle$ are trivial.

and (i) $\langle \{[0], [2], [4]\}, +_6 \rangle = H_1$ is a subgroup of $\langle \mathbb{Z}_6, +_6 \rangle$

(ii) $\langle \{[0], [3]\}, +_6 \rangle = H_2$ is a subgroup of $\langle \mathbb{Z}_6, +_6 \rangle$

H_1 & H_2 are proper subgroups of $\langle \mathbb{Z}_6, +_6 \rangle$

Note $H_1 = \langle [2] \rangle$ cyclic group generated by $[2]$ under $+_6$

$H_2 = \langle [3] \rangle$ " " " " " "

$[3]$ under $+_6$

Note that 2, 3 are the divisors of 6.

In general,

Note that $\langle \mathbb{Z}_n, +_n \rangle$ is a cyclic group.

We can generate all the elements of $\mathbb{Z}_n$ by [1] and some other elements of $\mathbb{Z}_n$.

Also note if k divides n then the set generated by k under $+$ forms a subgroup of $\mathbb{Z}_n$.

Theorem 3 A subset $S \neq \emptyset$ of G is a subgroup of $\langle G, * \rangle$ iff for any $a, b \in S$, $a * b^{-1} \in S$.

Proof:- Given $S \neq \emptyset$ is subset of G . — ①

Assume: S is a subgroup of $\langle G, * \rangle$ — ②

To prove: $\forall a, b \in S \Rightarrow a * b^{-1} \in S$.

As, $a, b \in S \Rightarrow b^{-1} \in S$ by ②

$\therefore a, b^{-1} \in S$

Hence $a * b^{-1} \in S$ by ②

Assume: $\forall a, b \in S \Rightarrow a * b^{-1} \in S$. — ③

To prove: S is a subgroup of $\langle G, * \rangle$

(i) To prove $e \in S$.

by ③ $\forall a, b \in S \Rightarrow a * b^{-1} \in S$

consider $b = a$

$$\therefore a * a^{-1} = e \in S$$

(ii) Inverse:

by ③: $\forall a, b \in S \Rightarrow a * b^{-1} \in S$

$$a = e \text{ and } b = a$$

consider

$$\Rightarrow e * a^{-1} = a^{-1} \in S$$

(iii) closed:

by ③, $\forall a, b \in S \Rightarrow a * b^{-1} \in S$

Consider $\forall a, b^{-1} \in S$ by (ii)

$$\Rightarrow a * (b^{-1})^{-1} \in S \text{ by ③}$$

$$\Rightarrow a * b \in S$$

$\therefore S$ is a subgroup of $\langle G, * \rangle$.

Group Homomorphism:

Let $\langle G, * \rangle$ and $\langle H, \Delta \rangle$ be two groups.
A mapping $g: G \rightarrow H$ is called group
Homomorphism from $\langle G, * \rangle$ to $\langle H, \Delta \rangle$ if
for any $a, b \in G$
 $g(a * b) = g(a) \Delta g(b)$.

Thm: 4 Group homomorphism preserves the identity, inverses and subgroups.

ie, (i) $g(e_G) = e_H$

(ii) $g(a^{-1}) = g(a)^{-1}$

(iii) If $(S, *)$ be a subgroup of $(G, *)$ then $(g(S), \Delta)$ be a subgroup of (H, Δ) .

Proof: Refer page 330 & 331 of J.P. Tremblay & Manohar.

Defn: Let g be a group homomorphism between $(G, *)$ to (H, Δ) . The set of all elements of G which are mapped into e_H , the identity elt of H is called the kernel of the homomorphism g .
denote $\ker(g)$.

ie, $\ker(g) = \{a \in G : g(a) = e_H\}$

Thm: 5 Prove that the kernel of a homomorphism g from a group $(G, *)$ to (H, Δ) forms a subgroup of $(G, *)$.

Proof: Note $g(e_G) = e_H$ by Thm 4

$\therefore e_G \in \ker(g)$

$\therefore \ker(g) \neq \phi$.

$\forall a, b \in \ker(g)$ to prove $a * b^{-1} \in \ker(g)$
 as $a, b \in \ker(g) \Rightarrow g(a) = e_H$ & $g(b) = e_H$ (*)

$$\begin{aligned} \text{consider } g(a * b^{-1}) &= g(a) \Delta g(b^{-1}) \\ &\quad \text{[by } g \text{ is homo]} \\ &= g(a) \Delta [g(b)]^{-1} \quad \text{[by Thm 4]} \\ &= e_H \Delta (e_H)^{-1} \quad \text{by (*)} \\ &= e_H \Delta e_H \end{aligned}$$

$$g(a * b^{-1}) = e_H$$

$$\Rightarrow a * b^{-1} \in \ker(g)$$

$\therefore \ker(g)$ is a subgroup of $(G, *)$.

Thm: 6 : show that every cyclic group of order n is isomorphic to the group $\langle \mathbb{Z}_n, + \rangle$.

proof:

Given $(G, *)$ be cyclic group of order n

$$\text{Let } G = \{a, a^2, a^3, \dots, a^{n-1}, a^n = e\}$$

where a is the generator of G .

$$\text{Let } \mathbb{Z}_n = \{[0], [1], [2], \dots, [n-1]\}$$

$$\text{Define : } g : (G, *) \rightarrow (\mathbb{Z}_n, +)$$

$$\text{by } g(a^j) = [j] \quad \text{for } j = 0, 1, 2, \dots, n-1.$$

note that $[1]$ is the generator of $(\mathbb{Z}_n, +)$
 and $g([1]) = a$ is the generator of $(G, *)$

It is easy to check g forms an
 isomorphism from $(G, *)$ to $(\mathbb{Z}_n, +)$.

Thm 7: Every finite group of order n is
 isomorphic to a permutation group
 of order n .

Proof:

Given G - be finite group of order n . w.r to $*$
 $\forall a \in G$
 Define a mapping $f_a: G \rightarrow G$ by $f_a(x) = a * x$
 [by simplicity we take $f_a(x) = ax$]

Claim: f_a is permutation on G $\forall a \in G$.
 its enough to prove f_a is 1-1 & onto

f_a is one-one:

consider $f_a(x) = f_a(y)$

by defn: $ax = ay$

$\Rightarrow x = y$ [by LCL]

$\therefore a$ here is not 0

$\therefore f_a$ is one-one

f_a is onto:

$\forall b \in G$ then there exist
 $x = a^{-1}b$ in G s.t

$f_a(x) = ax = a a^{-1}b = b$

$\therefore f_a$ is onto.

$x \in G$, f_a is bijective $\Rightarrow f_a$ is permutation.
Hence claim 1

consider $G' = \{ f_a : a \in G \}$
As $|G| = n \Rightarrow |G'| = n$

claim 2 G' is group w.r to composition $\uparrow$

(i) closure: for any $f_a, f_b \in G'$

$$\begin{aligned} f_a \circ f_b(x) &= f_a(f_b(x)) = f_a(bx) = abx \\ &= f_{ab}(x) \in G' \quad \because ab \in G. \end{aligned}$$

$\therefore G'$ is closed w.r to $\uparrow$

(ii) Associative is trivial for composition.

(iii) Identity:

for $e \in G$, we have $f_e \in G'$

$$\text{by defn } f_e(x) = ex = x$$

$\therefore f_e$ is an identity permutation

$\therefore f_e$ is identity elt in G'

(iv) Inverse:

for $a \in G$, $f_a \in G'$ note $a^{-1} \in G \Rightarrow f_{a^{-1}} \in G'$

$$\text{consider } f_a \circ f_{a^{-1}} = f_{aa^{-1}} = f_e$$

$$\text{or } f_{a^{-1}} \circ f_a = f_{a^{-1}a} = f_e$$

$\therefore f_{a^{-1}}$ is an inverse of f_a in G'

Inverse also true

Hence the claim 2.

$(G', \circ)$ is permutation group.

Define $\phi : G \rightarrow G'$ by
 $\phi(a) = f_a \quad \forall a \in G$

Claim: ϕ is isomorphism.

(i) ϕ is one-one.

consider $\phi(a) = \phi(b)$

by defn $\Rightarrow f_a = f_b$

$$\Rightarrow f_a(x) = f_b(x)$$

$$\Rightarrow ax = bx \Rightarrow a = b$$

$\therefore \phi$ is one-one

(ii) ϕ is onto.

by defn ϕ is onto.

[$\because \forall f_a \in G' \exists a \in G$ s.t. $\phi(a) = f_a$]

(iii) ϕ is homomorphism

consider $\phi(a * b) = f_{a * b} = f_{ab}$

$$= f_a \circ f_b$$

$$= \phi(a) \Delta \phi(b)$$

$\therefore \phi$ is homo

$\therefore \phi$ is isomorphism

$$\therefore G \cong G'.$$

Ex: 1 Consider $G = \{1, -1, i, -i\}$,
 define $f_a : G \rightarrow G$ by $f_a(x) = ax \quad \forall x \in G$.

$$f_1 : \begin{cases} f_1(1) = 1 \\ f_1(i) = i \\ f_1(-1) = -1 \\ f_1(-i) = -i \end{cases} \quad \begin{pmatrix} 1 & -1 & i & -i \\ 1 & -1 & i & -i \end{pmatrix}$$

$$f_{-1} : \begin{pmatrix} 1 & -1 & i & -i \\ -1 & 1 & -i & i \end{pmatrix}$$

$$f_i : \begin{pmatrix} 1 & -1 & i & -i \\ i & -i & -1 & 1 \end{pmatrix}$$

$$f_{-i} : \begin{pmatrix} 1 & -1 & i & -i \\ -i & i & 1 & -1 \end{pmatrix}$$

$$G' = \{f_1, f_{-1}, f_i, f_{-i}\}$$

then $\phi(1) = f_1, \phi(-1) = f_{-1}, \phi(i) = f_i, \phi(-i) = f_{-i}$

$$G \cong G'$$

Ex: 2 homomorphism

Ex: 2 $g : \mathbb{Z}_4 \rightarrow \mathbb{B}$

$$\mathbb{B} = \{0, 1\}$$

$$g([0]) = g([2]) = 0, \quad g([1]) = g([3]) = 1.$$

Ex: 3 $g : S \rightarrow P$

$$g(a) = 3, \quad g(b) = 1, \quad g(c) = 2$$

$$\begin{aligned} g : \mathbb{Z}_5^+ &\rightarrow \mathbb{Z}_5^+, * \mapsto * \\ g(0) &= 1, g(1) = 2 \\ g(2) &= 4, g(3) = 3 \\ g &\text{ is h.m.} \end{aligned}$$

by

$*$	a	b	c
a	a	b	c
b	b	b	c
c	c	b	c

$\oplus$	1	2	3
1	1	2	1
2	1	2	2
3	1	2	3

$$g(a * a) = g(a) \oplus g(a) \Rightarrow g(a) = 3 \oplus 3 = 3$$

$$g(a * b) = g(a) \oplus g(b) \Rightarrow g(b) = 3 \oplus 1 = 1$$

Coset

Let $(G, *)$ be a group and $(H, *)$ be a subgroup of G .

A left coset is an equivalence relation with respect to subgroup $(H, *)$ in $(G, *)$.

it is also denoted by modulo H .

ie for any $a \in G$, the left coset relation is

$$\begin{aligned} \text{defined as } [a] &= \{x \in G : x \equiv a \pmod{H}\} \\ &= \{x \in G : a^{-1} * x \in H\} \\ &= \{a * h : h \in H\} \\ &= a * H \text{ (or) simply } aH. \end{aligned}$$

Defn: Let $(H, *)$ be a subgroup of $(G, *)$.

For any $a \in G$, the set $a * H$ (or) aH defined by $aH = \{a * h : h \in H\}$ called the left coset of H in G .

||| eg we define right coset of H in G by

$$Ha = \{h * a : h \in H\}$$

Observation: The set of left coset of H in G forms a partition in G .

ie., Every element belongs to only one left coset of H in G .

Example: Consider $(\mathbb{Z}_6, +_6)$ be a group. $= G$

let $H = \{[0], [2], [4]\}$, $+_6$ be a subgroup of G .

then $\forall a \in G$, we define left coset and Right coset Resp. under $+_6$.

$$[0] +_6 H = \{[0], [2], [4]\}$$

$$[1] +_6 H = \{[1], [3], [5]\}$$

$$[2] +_6 H = \{[2], [4], [0]\}$$

$$[3] +_6 H = \{[3], [5], [1]\}$$

$$[4] +_6 H = \{[4], [0], [2]\}$$

$$[5] +_6 H = \{[5], [1], [3]\}$$

$$H + [0] = \{[0], [2], [4]\}$$

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For the subgroup: $H = \{[0], [3]\}$ in G .

The left coset:

$$[0] +_6 H = \{[0], [3]\}$$

$$[1] +_6 H = \{[1], [4]\}$$

$$[2] +_6 H = \{[2], [5]\}$$

$$[3] +_6 H = \{[3], [0]\}$$

$$[4] +_6 H = \{[4], [1]\}$$

$$[5] +_6 H = \{[5], [2]\}$$

(2) For the subgroup $\{[0], [2]\}$ of $(\mathbb{Z}_4, +_4)$
Find all the distinct left coset of H in G .

(3) Find the set of all distinct Right coset of
 $\left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \right\}$ in $(S_3, \cdot)$.

Lemma 1: Any two left (right) coset of H in G are either identical (or) disjoint.

Proof: Let H be a subgroup of $(G, *)$
consider two left coset aH & bH of H in G
for $a, b \in G$.

We need to prove $aH = bH$ (or) $aH \cap bH = \emptyset$.

Suppose not, if $aH \cap bH \neq \emptyset$

Then $\exists x \in aH \cap bH$

$\Rightarrow x \in aH$ and $x \in bH$

$\Rightarrow x = a * h_1$ for some $h_1 \in H$
& $x = b * h_2$ for some $h_2 \in H$

$\Rightarrow a * h_1 = b * h_2$ for some $h_1, h_2 \in H$

multiply h_1^{-1} on R. side.

$$a * h_1 * h_1^{-1} = b * h_2 * h_1^{-1}$$

$$\Rightarrow a = b * (h_2 * h_1^{-1})$$

$$a = b * h_3 \quad \text{for some } h_3 = h_2 * h_1^{-1}$$

$$\therefore aH = b * h_3 * H = bH \quad \because h * H = H.$$

Hence proved.

Lemma 2: For a subgroup H in G , $o(H) = o(aH)$ for any $a \in G$.

proof:

Given H is a subgroup of $(G, *)$.

Let $o(H) = m$: $H = \{h_1, h_2, \dots, h_m\}$

Now to prove $o(aH) = m$ for any $a \in G$.

suppose, not $o(aH) \neq m \Rightarrow o(aH) < m$.

i.e. for some $h_1, h_2 \in H$, $a * h_1 = a * h_2$ distinct.

but on $a \in G$, $a^{-1} \in G$.

$\therefore a^{-1} * a * h_1 = a^{-1} * a * h_2 \Rightarrow h_1 = h_2$

Hence the contradiction.

$\Rightarrow o(H) = o(aH)$ for any $a \in G$.

Theorem [Lagrange's theorem]

STATEMENT: The order of a subgroup of a finite group divides the order of the group.

Proof:-

Let $(G, *)$ be a finite group of order n and let $(H, *)$ be its subgroup.

Case 1: If H is either $\{e\}$ or G then the theorem is trivial $\therefore 1/n + n/n$.

If H is a proper subgroup.

Let $o(H) = m$ where $m < n$.

Let $H = \{h_1, h_2, \dots, h_m\}$.

Define a left coset relation w.r to H in G
as for $a, b \in G$, $a \equiv b \pmod{H}$ if $b^{-1} * a \in H$.

Claim 1: left coset relation is an equivalence relation.

(i) Reflexive: for $a \in G$,

$$\because a^{-1} * a = e \in H \because H \text{ is subgroup}$$

$$\Rightarrow a \equiv a \pmod{H}$$

(ii) Symmetry:

$$\text{for } a, b \in G, \text{ if } a \equiv b \pmod{H}$$

$$\Rightarrow b^{-1} * a \in H$$

$$\text{As } H \text{ is a subgroup} \Rightarrow (b^{-1} * a)^{-1} \in H$$

$$\Rightarrow a^{-1} * (b^{-1})^{-1} \in H$$

$$\Rightarrow a^{-1} * b \in H$$

$$\Rightarrow b \equiv a \pmod{H}$$

(iii) transitive:
for $a, b, c \in G$, if $a \equiv b \pmod{H}$ & $b \equiv c \pmod{H}$
Then $b^{-1} * a \in H$ & $c^{-1} * b \in H$

$\Rightarrow H$ is closed $\Rightarrow c^{-1} * b * b^{-1} * a \in H$

$\Rightarrow c^{-1} * a \in H$

$\Rightarrow a \equiv c \pmod{H}$

$\therefore$ left coset relation is an equivalence relation.

Define a left coset for every $a \in G$ as
 $[a] = \{a * h : h \in H\} = a * H$

Hence by equivalence relation, G will be partitioned into left cosets.

By Lemma 2: Any left cosets of H in G either identical or disjoint.

Hence $G = a_1 H \cup a_2 H \cup \dots \cup a_k H$
[let there are k distinct left cosets of H in G]

$$\Rightarrow |G| = |a_1 H| + |a_2 H| + \dots + |a_k H|$$

by lemma 2: $|H| = |aH|$ for any $a \in G$.

$$\therefore |G| = |H| + \dots + |H| \text{ (k times)}$$

$$n = mk \quad \therefore |H| = m$$

$$\Rightarrow m \mid n$$

Hence the theorem.

Converse of Lagrange's Thm. is false.

That if $m \mid n$ where $|G| = n$
then it is not necessary that
 G must have a subgroup of order m .

For Example:

Let $G = A_4$: Alternating group.
{ set of all even permutations on 4 elements }
 $|A_4| = 12$

Here 6 is divisor of 12, but in A_4 there
is no subgroup of order 6.

Note: From Lagrange's Thm:- if $|G| = n$
and if m does not divide n then
there is no subgroup of order m in G .

Group Code:

Most of the systems today contain telephone and communication lines which cause transmitted message to be corrupted by the presence of noise.

Structure in design of error-correcting codes is important.

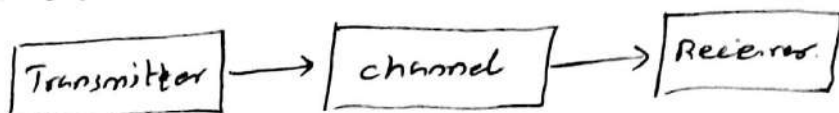
Even though the sender may like to have his message received by the receiver without any distortion, it is not possible due to a variety of disturbances (noise) to which the communication channel is subjected. Coding theory deals with minimizing the distortions of the conveyed message (due to noise) and retrieve the original message to the optimal extent possible from the corrupted message.

Encoder:- A device that can be used to improve the efficiency of the communication channel. It will transform the incoming messages in such a way that the presence of noise on the transformed messages is detectable.

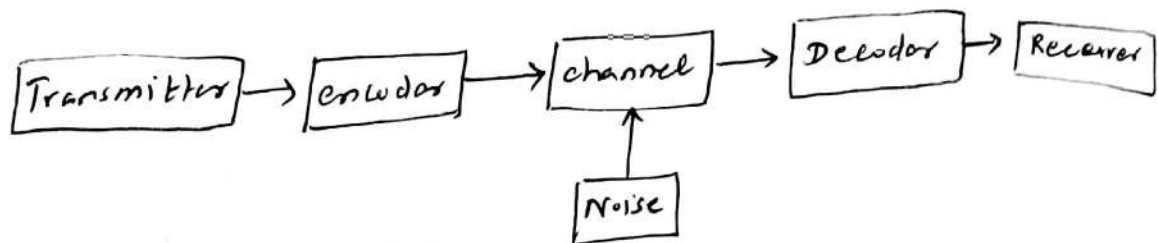
Decoder:- A decoder is a device which transforms the encoded message into their original form that can be understood by the receiver.

By using suitable encoder and decoder, it may be possible to detect the distortions in the messages due to noise in the channel and to

An ideal communication system



can now be modified to include the encoder & decoder, and to show the presence of noise in the channel.



Note:

- (1) The input message can consist of a sequence of letters, characters or symbol set called alphabet.
- (2) The input message will be transformed by the encoder into a string of characters or symbols of another alphabet in a one-one fashion.
- (3) We will discuss a binary channel in which the encoder will transform an input message into a binary string consisting of the symbols 0 (or) 1.
- (4) Decoding can be seen as the inverse operation of encoding.

Group Code:

If $B = \{0, 1\} \cong \mathbb{Z}_2$ under $+$ then

$B^n = \{x_1, x_2, x_3, \dots, x_n : x_i \in B\}$ is a group under binary operation of addition modulo 2, denoted $\oplus$, then $(B^n, \oplus)$ is called group code.

Note: As each binary sequence (code) consist of n binary digit, we have 2^n binary codes are possible.

Hamming distance: Suppose x and y represent two binary strings of length n binary $x_1, x_2, \dots, x_n$ & $y_1, y_2, \dots, y_n$ then the change in the number of position in the string x & y i.e. $x_i \neq y_i$ is called Hamming distance.

Denoted by $H(x, y) = \text{weight of } x \oplus y = |x \oplus y|$

or $H(x, y) = \sum_{i=1}^n (x_i \bar{y}_i)$ where $\begin{array}{c|cc} \bar{y} & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array}$

Example: if $x, y, z \in B^4$: $x = 1001$, $y = 0100$, $z = 1000$
Then $H(x, y) = 3$, $H(x, z) = 1$, $H(y, z) = 2$.

Generator matrix: - in Encoding

Suppose we have an encoding function $E: B^m \rightarrow B^n$ where $m < n$. [That is it transform the m bit information (i) to an n bit code].

The matrix form $[I_{m \times m} : A_{m \times n-m}] = G$ called the generator matrix for E .
Where $I_{m \times m}$ = Identity matrix of order m .

A is any matrix of entries from B .
[A - called parity matrix]
The code word is given by the matrix multiplication, modulo-2, of Row matrix corresponding to the message and G .

i.e., if m is the message

Then code word: $C = M \cdot G$

Example: Given $E: B^3 \rightarrow B^6$ by

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} \cong \begin{bmatrix} I_3 & A_{3 \times 3} \end{bmatrix}$$

$$m=3, n=6.$$

Consider a code (message):

$$m_1 = 011 \quad C_1 = m_1 \cdot G = (011) \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix} \\ = (011010) = e(011)$$

$$m_2 = 101 : C_1 = m_2 \cdot G = (101011) = e(101)$$

Similarly we can generate code word for all element in B^3 .

m_1	0 0 0	C_1	: 0 0 0 0 0 0
m_2	0 0 1	C_2	: 0 0 1 1 0 1
m_3	0 1 0	C_3	: 0 1 0 0 1 1
m_4	0 1 1	C_4	: 0 1 1 1 1 0
m_5	1 0 0	C_5	: 1 0 0 1 1 0
m_6	1 0 1	C_6	: 1 0 1 0 1 1
m_7	1 1 0	C_7	: 1 1 0 1 0 1
m_8	1 1 1	C_8	: 1 1 1 0 0 0

Note: Each code word C contains the original message (The identity matrix will fix the original message) and the parity bits (will be generated by the parity matrix A).

$$\text{ie, } C = [m : P]$$

$$\text{ie, } m \cdot G = [m : P] \\ m \cdot [I : A] = [m : P] \quad \text{where } P = m \cdot A$$

- * We call it as linear error correcting code.
we also have another called Hamming code.

Dedding: Parity check matrix:

The codeword (Hamming code) of a original message generated from the generator matrix can be deduced using the parity check matrix. Denoted by H .

Given $E: B^m \rightarrow B^n$ by $G = [I_{m \times m} : A_{m \times n-m}]$

The parity check matrix $H = [A^T : I_{n-m}]$

- * This matrix is used at the receiver side to detect the error.

Example:

Given generator matrix:

$$\text{Let } G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

generate parity check matrix for (7,4) code.

Given $G = [I_4 : A]$

$$n=7, m=4$$

$$H = [A^T : I_{7-4}]$$

$$H = [A^T : I_3]$$

$$H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Error Detection and Correction using parity check matrix

If c is a valid code word generated from the generated matrix G then $c \cdot H^T = \vec{0}$
(or) $H \cdot c^T = \vec{0}$.

Example: Let $m = (0101)$ be a original message
Use $e: B^4 \rightarrow B^7$ by $G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = [I_4 : A]$

Then the code word $c = m \cdot G = (0101100)$

To check c is a valid code: (no error)

check: $H \cdot c^T = \vec{0}$

Here $H = [A^T : I_3] = \begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$

$$\text{Hence } H \cdot c^T = \begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Example: Check $c = (1001100)$ is the valid code or not using $H = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$.

$$\text{check: } H \cdot c^T = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \neq \vec{0}$$

$\therefore$ The given code word have some error.

Detection of error: If $H \cdot c^T \neq \vec{0}$.

Let $H \cdot c^T = s$ where $s \neq \vec{0}$ called syndrome vector.

$$\text{ie, } H \cdot (\vec{c}_{\text{err}} + \vec{e}_r)^T = H \vec{c}_{\text{err}}^T + H \vec{e}_r^T \\ = \vec{0} + \vec{s}$$

if $\vec{e}_r = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ then the syndrome vector is in i^{th} column of H .

Which says that the code word c contains the error in i^{th} place.

For example: Let $\vec{c} = (1001100)$ (wrong)

$$\vec{c}_{\text{err}} = (1001100)$$

$$\vec{e}_r = (0100000)$$

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$$\text{Hence } H \vec{c}^T = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = s$$

Note that s is the 2nd column of H .

$$\text{Also } \begin{pmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} (\vec{c}_{\text{err}} + \vec{e}_r) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\Rightarrow H \vec{c}_{\text{err}} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, H \vec{e}_r = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

The error in c is at 2nd position.

$\therefore$ The correct code word for c is

$$c = (1001100)$$

problem: An encoding function $e: B^3 \rightarrow B^5$ is given by the parity check matrix is

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

(i) Determine all code words of B^3

(ii) Decode the received words

11101, 11011, 11110

Ans:

Given:

$$H = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix} \approx (A : I_3)$$

$n=5, m=3.$



The corresponding generator matrix:

$$G = (I_2 : A^T) = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}$$

(i) All the code word for B^2

For a message: $m \in B^2$

The code word $c = m \cdot G$

$$m_1 = 00 : e(00) = (00) \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix} \\ = (00000)$$

$$m_2 = 01 : e(01) = (01) \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix} \\ = (01101)$$

$$m_3 = 10 : e(10) = (10010)$$

$$m_4 = 11 : e(11) = (11111)$$

(ii) The received code:

$$c_1 = (11101), c_2 = (11011), c_3 = (11110)$$

To check the received code is correct.

$$\text{Then } H \cdot c^T = \vec{0}$$

$$\text{For } c_1 = (11101) : \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\therefore c_1$ is correct code (valid code)

$\therefore$ The correct message $m_1 = 11$

For $C_2 = (11011)$: Check $H \cdot C_2^T = \vec{0}$

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \text{syndrome vector}$$

$\therefore$ The given received message C_2 contains some error. $\checkmark$

Since syndrome vector $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is the first

column of H . $\therefore$ The corrected C_2

$$\text{is } C_2' = (01011)$$

$\therefore$ The corrected message is $m_2: (01)$

For $C_3 = (11110)$

Check: $H \cdot C_3^T = \vec{0}$

$$\begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \text{Syndrome vector}$$

as $\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ is coincide with column 2 of H .

$\therefore C_3$ have error in 2nd position

$$\therefore C_3' = (10110)$$

$\therefore$ The corrected message: (10) .

13) Find the code generated by the given parity matrix 'H' when the encoding function is $e: B^3 \rightarrow B^6$, where

$$H = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$$

To find the code word C s.t.

$$H \cdot C^T = \vec{0}$$

$$\begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \\ c_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{aligned} C &= (111000), & C &= (100110), & C &= (011110) \\ (3) \quad C &= (001011), & C &= (000000) \\ C &= (010101), & C &= (101101) \\ & & C &= (110011) \end{aligned}$$